Exercise list R8b — Executive compensation and managerial incentives

Covers "remuneração de executivos" from the course ementário: why managers want growth, pay levels versus pay-for-performance, bonuses on accounting profit, stock and option grants, the risk-sharing versus incentives trade-off in the optimal fee schedule, the power of the incentive scheme, and golden parachutes. It complements R8 (governance and control rights) and R4 (the moral-hazard financing model). 15 real exercises ordered easy → really hard, copied verbatim from the course books for personal study; pages are printed, with the PDF page in brackets. The pool had 17 unused compensation exercises; at the cap of 15, Tirole 7.5 (benchmarking, relative performance with risk-averse entrepreneurs) and Amaro de Matos 3.1.5 #1 (managerial ownership and control in the §3.1.1 model) were left out. Worked solutions: `lista-r8b-executive-compensation-solutions.pdf` (short answers: `...-solutions-short.pdf`). Links: exercise index · course programme.

References

#	Tier	Book	Ch.	Exercise	Printed p. (PDF)	Answer in book	Trains
1	Easy	Ross *Fundamentos* 9e	Cap. 1	CQ 1.16	23 (47)	no	CEO p
2	Easy	Berk–DeMarzo–Harford 2e	Ch. 22	Problem 3	677 (708)	no	A 3% C
3	Easy	Grinblatt & Titman 2e	Ch. 18	18.1	653 (666)	no	Why m
4	Easy	Higgins (Koski–Higgins–Mitton) 13e	Ch. 1	1.2	32 (53)	no	A bonu
5	Easy	Grinblatt & Titman 2e	Ch. 18	18.7	653 (666)	no	Stock-b
6	Exam	Grinblatt & Titman 2e	Ch. 18	18.9	654 (667)	no	Pay on
7	Exam	Grinblatt & Titman 2e	Ch. 18	18.6	653 (666)	no	Chrysle
8	Exam	Grinblatt & Titman 2e	Ch. 18	18.3	653 (666)	no	Equity
9	Exam	Grinblatt & Titman 2e	Ch. 18	18.8	653–654 (666)	no	Leverag
10	Exam	Amaro de Matos	Ch. 3	3.1.5 #3	77 (95)	yes, p. 256	Stock o
11	Exam	Amaro de Matos	Ch. 3	3.1.5 #4	77 (95)	yes, pp. 256–257	Pareto-
12	Really hard	Amaro de Matos	Ch. 3	3.1.5 #6	77 (95)	yes, p. 257	CARA
13	Really hard	Tirole	Ch. 3	3.10	148 (161)	yes, p. 575	Congru
14	Really hard	Amaro de Matos	Ch. 3	3.1.5 #5	77 (95)	yes, p. 257	Second
15	Really hard	Tirole	Ch. 7	7.9	326 (339)	yes, pp. 604–605	Optima

Tier 1 — Easy

1. Is CEO pay too high?

Source: Ross, Cap. 1, Questão 16 (Revisão de conceitos e questões instigantes), p. 23 (PDF 47) · Answer in book: no

16. Remuneração dos executivos [OA3] Os analistas têm indicado que a remuneração dos administradores de primeiro escalão nos Estados Unidos é alta demais e deve ser reduzida. Por exemplo, concentrando-nos nas grandes corporações, Ray Irani, da Occidental Petroleum, é um dos CEOs mais bem pagos, tendo ganhado cerca de $54,4 milhões em 2007 e $550 milhões no período de 2003 a 2007. Tais valores são excessivos? Ao responder essa pergunta, seria importante reconhecer que superestrelas dos esportes, como Tiger Woods, grandes celebridades, como Tom Hanks e Oprah Winfrey, e muitos outros profissionais de sucesso em suas respectivas áreas ganham tanto quanto ou mais.

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2. A small CEO stake and a value-destroying acquisition

Source: Berk, Ch. 22, Problem 3, p. 677 (PDF 708) · Answer in book: no

3. You are invested in GreenFrame, Inc. The CEO owns 3% of GreenFrame and is considering an acquisition. If the acquisition destroys $50 million of GreenFrame's value, but the present value of the CEO's compensation increases by $5 million, will he be better or worse off?

Context needed to solve it *From Berk §22.3 "Reasons to Acquire", "Managerial Motives to Merge", p. 659 (PDF 690):*

Conflicts of Interest. Managers may prefer to run a larger company due to the additional pay and prestige it brings. Because most CEOs hold only a small fraction of their firm's stock, they may not bear enough of the cost of an otherwise bad merger that increases their personal benefits. For example, a CEO who owns 1% of her firm's stock bears 1% of every dollar lost on a bad acquisition, but enjoys 100% of the gains in compensation and prestige that come with being the CEO of a larger company. If the acquisition destroys $100 million in shareholder value, but increases the present value of her compensation by more than $1 million, she will prefer to execute the merger anyway. Why would the board of directors create these incentives? Either due to poor monitoring of the manager, or belief that the strategy is correct even if the stock market disagrees, boards typically increase the pay of CEOs along with the size of the firm, even if the size comes at the expense of poorly performing acquisitions.

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3. Why managers want growth

Source: Grinblatt, Ch. 18, Exercise 18.1, p. 653 (PDF 666) · Answer in book: no

18.1. Discuss why managers might tend to want their organizations to grow.

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4. A bonus after a boom year

Source: Higgins, Ch. 1, Exercise 1.2, p. 32 (PDF 53) · Answer in book: no (even-numbered; answers only in the instructor resources)

2. Braintree Corporation, a biotechnology company, has $7 billion in assets, $5 billion in equity, and earned a profit of $100 million last year as the economy boomed. Senior management proposes paying themselves a large cash bonus in recognition of their performance. As a member of Braintree's board of directors, how would you respond to this proposal?

Context needed to solve it *From Higgins Ch. 1, "Economic Income vs. Accounting Income — Imputed Costs", pp. 27–29 (PDF 48–50):*

A similar but subtler problem exists on the cost side of the income statement. It involves the cost of equity capital. Polaris's accountants acknowledge that in 2020 the company had use of $1,434 million of shareholders' money, measured at book value. They would further acknowledge that Polaris could not have operated without this money and that this money is not free. Just as creditors earn interest on loans, equity investors expect a return on their investments. […]

[…] Observe that while the accountant shows earnings of $414 million, the economist records a profit of only $271 million. These numbers differ because the economist includes a $143 million charge as a cost of equity capital, while the accountant pretends equity is free. (We will consider ways to estimate a company's cost of equity capital in Chapter 8. Here, for illustrative purposes only, I have assumed a 10 percent annual equity cost and applied it to the book value of Polaris's equity [$143 million = 10% × $1,434 million].)

($ millions)	Accountant	Economist
Operating income	$592	$592
Interest expense	67	67
Other nonoperating expenses	4	4
Cost of equity		143
Income before taxes	521	378
Provision for taxes	107	107
Accounting earnings	$414	
Economic earnings		$271

The distinction between accounting earnings and economic earnings might be only a curiosity if everyone understood that positive accounting earnings are not necessarily a sign of superior or even commendable performance. But

when many labor unions and politicians view accounting profits as evidence that a company can afford higher wages, higher taxes, or more onerous regulation, and when most managements view such profits as justification for distributing handsome performance bonuses, the distinction can be an important one. [...] Remember too that a company is not shooting par unless its economic profits are zero or greater.

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5. Stock-based pay and market-to-book ratios

Source: Grinblatt, Ch. 18, Exercise 18.7, p. 653 (PDF 666) · Answer in book: no

18.7. The tendency of firms to use stock-based compensation is higher for firms with higher market-to-book ratios. Provide two explanations for this empirical observation.

Tier 2 — Exam level

6. Paying the Chicago Bulls' coach

Source: Grinblatt, Ch. 18, Exercise 18.9, p. 654 (PDF 667) · Answer in book: no

18.9. Suppose that you are designing the compensation contract for the Chicago Bulls' new coach. Two main alternatives are possible. In (a) you will design his bonus based on the total number of wins during the season and the team's success during the playoffs and will ignore any specific decisions made by the coach. In (b) you will consider the specific measures taken by the coach and, perhaps with the help of independent outside experts, will base the compensation on the quality of those decisions but will ignore the number of wins during the season. Explain the advantages and disadvantages of the two compensation contracts.

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7. CEO contracts at Chrysler and Chevron

Source: Grinblatt, Ch. 18, Exercise 18.6, p. 653 (PDF 666) · Answer in book: no

18.6. You are a member of the compensation committee of the board of directors for both Chrysler and Chevron. How should the compensation contracts for the CEOs of these two companies differ?

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8. High Tech: new equity or a partially owned subsidiary?

Source: Grinblatt, Ch. 18, Exercise 18.3, p. 653 (PDF 666) · Answer in book: no

18.3. John Jacobs, the CEO of High Tech Industries, owns 51 percent of the shares of his $50 million company. The firm is starting a new project that requires $25 million in new equity capital. Jacobs is considering two ways to fund the project. The first is to issue $25 million in new equity. The second is to form a partially owned subsidiary of High Tech, which would be called Super Tech, and have the subsidiary issue the equity. Under the second proposal, Super Tech would be 55 percent owned by High Tech and 45 percent owned by new shareholders. Describe how the incentives of the managers of the new business and John Jacobs are likely to be affected by the two proposals.

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9. Cybertex's leveraged recapitalization

Source: Grinblatt, Ch. 18, Exercise 18.8, pp. 653–654 (PDF 666) · Answer in book: no

18.8. Cybertex's management currently owns 1 percent of the firm's outstanding shares. The firm is currently financed with 50 percent debt and 50 percent equity but is planning to increase its leverage ratio to 80 percent debt by borrowing and using the funds to repurchase shares. Management has decided not to participate in the repurchase so their percentage ownership of the firm will increase.

Explain how managers' investment incentives are likely to change after the recapitalization. Specifically, discuss their incentives to take:

- Negative NPV projects that benefit them personally.
- Risky projects.
- Long-term projects that take more than 10 years to provide an adequate return to capital.

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10. Stock options in the Jensen–Meckling model

Source: Amaro de Matos, Ch. 3, Exercise 3.1.5 #3, p. 77 (PDF 95) · Answer in book: yes, p. 256

3. Some authors argue that one way to control the conflicts of interests between managers and stockholders is to give stock options to managers. How can we explain this argument using the simple model of section 3.1.1? What other factors should be included in this analysis?

Context needed to solve it *From Amaro de Matos §3.1.1 "Agency Costs of Outside Equity", pp. 64–66 (PDF 82–84):*

Consider a firm where the manager owns a fraction of the equity. This manager-owner will make operating decisions maximizing the joint effect of two different sources of utility. The manager's utility will be assumed to be generated not only by the benefits derived from pecuniary returns, but also by some nonpecuniary aspects of the manager's activity. Examples of nonpecuniary aspects are the physical appointments of the office, the kind and amount of contributions to charity, attractiveness of the staff, nature of personal relationship with employees, and an excellent computer.

When the owner-manager sells equity to outsiders, a typical agency problem arises because of the difference of interests between him/her and the outside shareholders.

Consider a firm generating a flow with present market value $\overline{V}$, and let F denote the market value of the stream of the manager's expenditures on nonpecuniary benefits. This reduces the effective value of the firm to $V = \overline{V} - F$. If the owner-manager holds a fraction α of the shares of the firm, its utility will be described by the real function

$$U(\alpha V, F),$$

which is increasing and concave in both arguments. In this context, two main results are in order. First, given an optimal choice of F utility to the owner-manager decreases as new equity is issued. Second, the effective value of the firm decreases. In fact, the decline in the value of the firm is entirely imposed on the owner-manager. This is reflected in the decrease of his/her utility.

For fixed α, the optimal value $F^*(\alpha)$ is obtained maximizing the utility

$$\max_F U[\alpha(\overline{V} - F), F].$$

In other words, $F^*(\alpha)$ must satisfy the first-order condition

$$-\alpha U_1 + U_2 = 0. \qquad (3.1)$$

Let $\xi(\alpha)$ denote the optimal value of the utility, or

$$\xi(\alpha) \equiv U[\alpha(\overline{V} - F^*(\alpha)), F^*(\alpha)].$$

Using the first-order condition (3.1), it is clear that $\xi(\alpha)$ is a monotonic increasing function of α, since

$$d\xi(\alpha) = (-\alpha U_1 + U_2)\, dF^* + V\, d\alpha = V\, d\alpha.$$

This means that as α increases, the optimal utility of the owner-manager increases. Issuing equity is just the reverse process, reducing the value of α. Thus, the first result comes:

Proposition 3.1.1. As new equity is issued, the utility to the owner-manager decreases.

It was just stressed that the decrease in the utility comes from the fact that the owner-manager carries all the reduction in value of the firm when new equity is issued. Indeed, since

$$V = \overline{V} - F,$$

it follows that $dV = -dF$. In order to show that the value increases with α, it suffices to show that F decreases. This follows since $dF/d\alpha$ can be obtained differentiating the first-order condition (3.1) and using the envelope theorem. It follows that

$$\frac{d^2U}{dF^2}\frac{dF}{d\alpha} = U_1.$$

By assumption $U_1 > 0$ and since these derivatives are calculated at the optimum,

$$\frac{d^2U}{dF^2} < 0,$$

leading to

$$\frac{dF}{d\alpha} < 0.$$

Thus, this confirms the second result:

Proposition 3.1.2. As new equity is issued, the effective value of the firm decreases.

It is clear from the preceding formalism that the reason why the value of the firm decreases is that the optimal market value F^* of the nonpecuniary benefits increases. The managers would have an incentive to convince the outsiders that the value of the firm will remain at the level $V^* = \overline{V} - F^*$ and the fringe benefits will be kept at F^*. Jensen and Meckling (1976) point out that this suggestion is not credible if the outsiders are rational. Basically, they know that it would be optimal for the management to stay at V^* only if they hold the fraction α of the shares.

Under the assumption that $\overline{V}$ is fixed, the new shareholders know beforehand that managers will be tempted to increase their benefits, given that now they only bear a smaller fraction of the costs associated with F. That is why selling some shares will decrease the value of the firm to a level $V^\circ < V^*$, with benefits at a level $F^\circ > F^*$. Knowing that, the new shareholders in fact would not be willing to pay more than the fraction of the new value V° corresponding to their fraction of shares.

In order to solve this problem, the management may accept being controlled by the outsiders. This can be done either by systematic auditing (to see whether $F = F^*$), by formal control systems, by budget restrictions, by incentive compensations to the management, or by the market for corporate control.

From Amaro de Matos §3.1.2, "Executive Stock Options", pp. 70–71 (PDF 88–89):

An interesting alternative to help solving the principal–agent problem in the context of management's compensation is to offer a package including call options on shares of the firm. Such options are typically offered *at the money*—that is to say, at an exercise price coinciding with the current value of the shares (say, S). Recalling that the payoff of such options at time T is

$$\max[0, S_T - S],$$

this component in an executive compensation package will give the right incentive for the manager to care about S_T, the future value of the shares.

Among other advantages, it is clear that the introduction of such instruments as a payment device align the incentives of managers with those of the shareholders. In fact, the residual claim to shareholders, as expressed in (2.2), can be seen as an option on the value of the firm with an exercise price that coincides with the face value of debt. If the management is given options, that constrains the real value of the payment to the performance of the firm, as reflected in the market value of the stock.

A second advantage is that this type of instrument allows the reduction of the base pay of the management, thus reducing salary differences among different employees.

Finally, in this setting, the payment of the manager is at risk. At least, this component of the package may be worthless at time T, which justifies a real commitment with the goals of the firm.

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11. The Pareto-efficient fee schedule

Source: Amaro de Matos, Ch. 3, Exercise 3.1.5 #4, p. 77 (PDF 95) · Answer in book: yes, pp. 256–257

4. Explain condition (3.4) and show what happens to the optimal value f*[w(a°, θ)] when θ changes.

Context needed to solve it *From Amaro de Matos §3.1.2 "Principal–Agent Problems", "Determining Optimal Fee Schedules", pp. 66–68 (PDF 84–86):*

The first formalization of the problem of determining an optimal fee schedule was made by Ross (1973). In this model the management takes an *unobservable* action that determines the probability distribution of the shareholders' payoff. The payment to the management is a function of the payoff, which is observable. The fee schedule is determined optimally by the shareholders in order to provide the incentives for the managers to choose the best possible actions when maximizing their utility. This maximization is constrained to providing managers a reservation expected utility.

Notice that if the actions of the management *are observable* and managers are risk averse, a first-best solution is attainable by making the fee very low, unless the optimal action (from the shareholders' viewpoint) is taken by the management. In other words, the payment schedule will share the risk associated with the managers' action optimally. If the actions are not observable, as in Ross's model, the expected utility for the shareholders will be lowered by their ignorance, since they will have to find a *costly* way to induce the managers to take an optimal action. The difference in the expected utility to the shareholders reflects the agency cost of monitoring the management through the payment schedule.

The problem is to define an optimal fee schedule f to the management such that it will be in the interest of the managers to take the actions desired by the shareholders.

Let a be the action chosen by the management, defined in a space $\mathcal{A}$ of feasible actions. Uncertainty is described by the set of states of nature Ω together with a probability distribution over the elements of Ω. As before, for $w \in \Omega$, the expectation of a random variable $x(w) \in R^{\Omega}$ on Ω is denoted by $E(x)$.

Let the result of action a be a payoff $\theta(a, w)$ in the state of nature $w \in \Omega$, which is assumed to be *unknown* beforehand to both management and shareholders. This payoff is thus a real valued function $\theta(a, w)$: $\mathcal{A} \times \Omega \to R$.

The fee f is assumed here to be a function *uniquely* of the payoff, that is,

$$f \equiv f(\theta(a, w)).$$

Managers and shareholders are supposed to have their preferences represented respectively by the utility functions G and U. The management faces the problem of choosing an optimal action a° as a function of a jointly optimal fee schedule f^* to be determined, such that

$$\max_{a \in \mathcal{A}} E\{G[f^*(\theta(a, w))]\}, \qquad (3.2)$$

constrained to a minimum level of expected utility k

$$E\{G[f^*(\theta(a^{\circ}, w))]\} > k.$$

Hence, the optimal action taken by the managers depends on the established fee schedule f^*. It will be denoted by $a(f^*)$. On the other hand, the shareholders will have to choose the optimal fee schedule f^* so it will be the schedule f satisfying the following variational optimization:

$$\max_f E\{U[\theta(a(f), w) - f(\theta(a(f), w))]\} \qquad (3.3)$$

subject to

$$E\{G[f(\theta(a^\circ, w))]\} > k.$$

The problem can be considerably simplified if one looks for a specific type of solution. Here, two kinds of fee schedules are proposed, following Ross's model.

1. A family of fee schedules can be achieved if managers and shareholders cooperate, maximizing a weighted sum of their expected utilities. In this case, f satisfies

   $$\max_f E\{U[\theta - f] + \lambda G[f]\}.$$

   where λ parameterizes the family, defining the weights of both utilities. This family of fees is said to be *Pareto-efficient*. When U and G are both monotonic and concave, the optimal fee schedule must maximize the argument of the expected value and therefore is defined implicitly by the equation

   $$U'[\theta - f] = \lambda G'[f]. \qquad (3.4)$$

2. An alternative approach to find a fee schedule is to assume that, at optimality, the utilities of managers and shareholders are positively and linearly related. That is to say, the optimal fee schedule must satisfy the *similarity condition*

   $$U[\theta - f] = cG[f] + b, \qquad (3.5)$$

   for constants c and b such that $c > 0$. It is clear that both managers and shareholders have the same attitude toward risky payoffs, and hence this similarity condition ensures that the managers will always choose the act more desired by the shareholders.

Ross argues that (3.4) and (3.5) are simultaneously satisfied only if both utility functions U and G are in the linear risk tolerance class. This will imply that the optimal fee schedule f is a *linear* function of the payoff.

Tier 3 — Really hard

12. CARA utilities and a linear fee schedule

Source: Amaro de Matos, Ch. 3, Exercise 3.1.5 #6, p. 77 (PDF 95) · Answer in book: yes, p. 257

6. Prove that when both utility functions V and G have constant absolute risk averse coefficients, the optimal fee schedule $f(\cdot)$ is a linear function of the payoff.

Context needed to solve it *From Amaro de Matos §3.1.2, "Determining Optimal Fee Schedules", pp. 67–68 (PDF 85–86):*

Let a be the action chosen by the management, defined in a space $\mathcal{A}$ of feasible actions. […] Let the result of action a be a payoff $\theta(a, w)$ in the state of nature $w \in \Omega$, which is assumed to be *unknown* beforehand to both management and shareholders. […]

The fee f is assumed here to be a function *uniquely* of the payoff, that is,

$$f \equiv f(\theta(a, w)).$$

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1. A family of fee schedules can be achieved if managers and shareholders cooperate, maximizing a weighted sum of their expected utilities. In this case, f satisfies

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13. Congruence and the power of the incentive scheme

Source: Tirole, Ch. 3, Exercise 3.10, p. 148 (PDF 161) · Answer in book: yes, p. 575

Exercise 3.10 (congruence, pledgeable income, and power of incentive scheme). The credit rationing model developed in this chapter assumes that the entrepreneur's and investors' interests are *a priori* dissonant, and that incentives must be aligned by giving the entrepreneur enough of a stake in the case of success.

Suppose that the entrepreneur and the investors have indeed dissonant preferences with probability x, but have naturally aligned interests with probability $1 - x$. Which prevails is unknown to both sides at the financing stage and is discovered (only) by the entrepreneur just before the moral-hazard stage.

More precisely, consider the fixed-investment model of Section 3.2. The investors's outlay is $I - A$ and they demand an expected rate of return equal to 0. The entrepreneur is risk neutral and protected by limited liability. With probability x, interests are dissonant: the entrepreneur obtains private benefit B by misbehaving (the probability of success is p_L) and 0 by behaving (probability of success p_H). With probability $1 - x$, interests are aligned: the entrepreneur's taking her private benefit B coincides with choosing probability of success p_H.

(i) Consider a "simple incentive scheme" in which the entrepreneur receives R_b in the case of success and 0 in the case of failure. R_b thus measures the "power of the incentive scheme."

Show that it may be optimal to choose a low-powered incentive scheme if preferences are rather congruent (x low) and that the incentive scheme is necessarily high-powered if preferences are rather dissonant (x high).

(ii) Show that one cannot improve on simple incentive schemes by presenting the entrepreneur with a menu of two options (two outcome-contingent incentive schemes) from which she will choose once she learns whether preferences are congruent or dissonant.

Context needed to solve it *From Tirole §3.2.1 "The Fixed-Investment Model", pp. 115–116 (PDF 128–129):*

Variants of the following entrepreneurial model will be used in the following: an entrepreneur (also called the "insider" or the "borrower," "she") has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. For the moment we interpret these assets as being cash or liquid securities that can be used toward covering the cost of investment. […] The entrepreneur's cash can either be invested in the project or used for consumption. To implement the project the entrepreneur must borrow $I - A$ from lenders. […]

Project. If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. The probability of success is denoted by p. The project is subject to moral hazard. The entrepreneur can "behave" ("work," "exert effort," "take no private benefit") or "misbehave" ("shirk," "take a private benefit") […].

Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. In the "effort interpretation," B can also be interpreted as a disutility of effort saved by the entrepreneur when shirking. Let $\Delta p \equiv p_H - p_L$.

Preferences and the loan agreement. Both the borrower and the potential lenders (or "investors") are risk neutral. For notational simplicity, there is no time preference; the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values.

Lenders behave competitively in the sense that the loan, if any, makes zero profit. [...]

Let us turn to the loan contract. A contract first stipulates whether the project is financed. If so, it further specifies how the profit is shared between the lenders and the borrower. The borrower's limited liability will imply that both sides receive 0 in the case of failure [...]. In the case of success, the two parties share the profit R; R_b goes to the borrower and R_l to the lenders. To sum up, we posit an incentive scheme for the entrepreneur of the following form: R_b in the case of success, 0 in the case of failure.

[...] We summarize the timing in Figure 3.1.

Figure 3.1 (timeline, left to right): Loan agreement (sharing rule) → Investment → Moral hazard → Outcome.

We assume that the project is viable only in the absence of moral hazard. That is, the project has positive NPV if the entrepreneur behaves,

$$p_H R - I > 0, \qquad (3.1)$$

but negative NPV, even if one includes the borrower's private benefit, if she does not,

$$p_L R - I + B < 0. \qquad (3.2)$$

From Tirole §3.2.2 "The Lenders' Credit Analysis", pp. 116–117 (PDF 129–130):

[...] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From this incentive compatibility constraint we infer that the highest income in the case of success that can be pledged to the lenders without jeopardizing the borrower's incentives is

$$R - \frac{B}{\Delta p}.$$

The *(expected) pledgeable income* is then

$$\mathcal{P} = p_H \left(R - \frac{B}{\Delta p} \right).$$

Because the lenders must break even in order to be willing to finance the project, a necessary condition for the borrower to receive a loan is that the expected pledgeable income exceed the lenders' initial outlay:

$$\mathcal{P} \equiv p_H \left(R - \frac{B}{\Delta p} \right) \geqslant I - A, \qquad (\text{IR}_l)$$

[...] Thus a necessary condition for financing to be arranged is

$$A \geqslant \bar{A} = p_H \frac{B}{\Delta p} - (p_H R - I). \qquad (3.3)$$

[…] Conversely, if $A \geqslant \bar{A}$, the entrepreneur is able to secure financing, and so condition (3.3) is both a necessary and a sufficient condition for financing. The entrepreneur offers claim R_l to competitive investors so as not to leave them with a surplus:

$$p_H R_l = I - A.$$

[…] Using the breakeven condition for the lenders ($p_H R_l = I - A$), the borrower obtains net utility or payoff (where "net" means that we subtract the consumption utility, A, that the entrepreneur would get by not undertaking the project):

$$U_b = \begin{cases} 0 & \text{if } A < \bar{A}, \\ p_H R_b - A = p_H(R - R_l) - A = p_H R - I & \text{if } A \geqslant \bar{A}. \end{cases} \qquad (3.6)$$

As could have been expected from the zero-profit condition for the lenders, *the borrower receives the entire social surplus or net present value if the project is funded.*

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14. Incentives versus efficient risk sharing

Source: Amaro de Matos, Ch. 3, Exercise 3.1.5 #5, p. 77 (PDF 95) · Answer in book: yes, p. 257

5. Show that the risk-sharing rule implied by condition U′/G′ = λ − ψ d(w_a/w_b)/dθ is not Pareto-efficient when the agent is risk averse. Is it when the agent is risk neutral?

Context needed to solve it *From Amaro de Matos §3.1.2 "Principal–Agent Problems", "Determining Optimal Fee Schedules", pp. 66–70 (PDF 84–88):*

The first formalization of the problem of determining an optimal fee schedule was made by Ross (1973). In this model the management takes an *unobservable* action that determines the probability distribution of the shareholders' payoff. The payment to the management is a function of the payoff, which is observable. The fee schedule is determined optimally by the shareholders in order to provide the incentives for the managers to choose the best possible actions when maximizing their utility. This maximization is constrained to providing managers a reservation expected utility.

Notice that if the actions of the management *are observable* and managers are risk averse, a first-best solution is attainable by making the fee very low, unless the optimal action (from the shareholders' viewpoint) is taken by the management. In other words, the payment schedule will share the risk associated with the managers' action optimally. If the actions are not observable, as in Ross's model, the expected utility for the shareholders will be lowered by their ignorance, since they will have to find a *costly* way to induce the managers to take an optimal action. The difference in the expected utility to the shareholders reflects the agency cost of monitoring the management through the payment schedule.

[…] Let a be the action chosen by the management, defined in a space $\mathcal{A}$ of feasible actions. Uncertainty is described by the set of states of nature Ω together with a probability distribution over the elements of Ω. […] Let the result of action a be a payoff $\theta(a, w)$ in the state of nature $w \in \Omega$, which is assumed to be *unknown* beforehand to both management and shareholders. This payoff is thus a real valued function $\theta(a, w)$: $\mathcal{A} \times \Omega \to R$.

The fee f is assumed here to be a function *uniquely* of the payoff, that is,

$$f \equiv f(\theta(a, w)).$$

Managers and shareholders are supposed to have their preferences represented respectively by the utility functions G and U. The management faces the problem of choosing an optimal action a° as a function of a jointly optimal fee schedule f^* to be determined, such that

$$\max_{a \in \mathcal{A}} E\{G[f^*(\theta(a, w))]\}, \qquad (3.2)$$

constrained to a minimum level of expected utility k

$$E\{G[f^*(\theta(a^\circ, w))]\} > k.$$

Hence, the optimal action taken by the managers depends on the established fee schedule f^*. It will be denoted by $a(f^*)$. On the other hand, the shareholders will have to choose the optimal fee schedule f^* so it will be the schedule f satisfying the following variational optimization:

$$\max_f E\{U[\theta(a(f), w) - f(\theta(a(f), w))]\} \qquad (3.3)$$

subject to

$$E\{G[f(\theta(a^\circ, w))]\} > k.$$

[...]

1. A family of fee schedules can be achieved if managers and shareholders cooperate, maximizing a weighted sum of their expected utilities. In this case, f satisfies

   $$\max_f E\{U[\theta - f] + \lambda G[f]\}.$$

   where λ parameterizes the family, defining the weights of both utilities. This family of fees is said to be *Pareto-efficient*. When U and G are both monotonic and concave, the optimal fee schedule must maximize the argument of the expected value and therefore is defined implicitly by the equation

   $$U'[\theta - f] = \lambda G'[f]. \qquad (3.4)$$

[...]

In the original problem, there is the restriction that the expected utility of the managers should not be below a certain level. Besides, there is a constraint that the managers will act so as to optimize their own expected utility, which is expressed by the first-order condition

$$E\{G'[f(\theta)]f'(\theta)\theta_a\} = 0, \qquad (3.6)$$

where θ_a denotes the derivative of θ with respect to a and $k = 0$. Then, the shareholders' problem is to maximize with respect to f the Lagrangean

$$\mathcal{L} = E\{U[\theta - f(\theta)] + \psi G'[f(\theta)]f'(\theta)\theta_a + \lambda G[f(\theta)]\},$$

where ψ and λ are the Lagrange multipliers associated respectively with the restrictions expressed in (3.6) and (3.3). Under some general conditions the Euler-Lagrange equation characterizes optimality provided that

$$\frac{U'}{G'} = \lambda - \psi \frac{d}{d\theta}\left(\frac{\theta_a}{\theta_w}\right).$$

Comparing this solution with (3.4), it follows that in general the fee schedule will be Pareto efficient if either

1. $\psi = 0$, meaning that (3.6) is not binding and the fee schedules will be Pareto-efficient independently of the payoff function θ, or
2. The term $d(\theta_a/\theta_w)/dw$ is a function only of a.

For the optimal fee schedule to be Pareto efficient independently of the payoff function $\theta(a, w)$, the first of these two conditions has to be fulfilled, implying a linear fee schedule, as shown by Ross, and therefore linear risk tolerance utility functions U and G.

[…]

Most of the subsequent literature in this topic is centered on the case where the managers are averse to effort and the shareholders try to maximize the value of the firm (i.e., are risk neutral). For instance, Shavell (1979) shows that under these conditions a Pareto optimal fee schedule does not maximize the expected profits of the shareholders. This, once again, reflects the costs of monitoring the actions of the management.

In principle, the risk associated with the output is shared among the management and the stockholders. The way this is done strongly depends on the risk-aversion characteristics of both shareholders and managers, as discussed by Shavell. Even in the case of risk neutrality of the shareholders, the fee schedule cannot be fixed as a constant value because then the managers would have no incentive to overcome their aversion to effort. The payment schedule has to depend on the output, leading as before to a second-best solution.

It may be proved that the shareholders will use any indication of the level of effort of the management in the equilibrium fee schedule, in order to approach its first-best solution. Information, even if incomplete, thus has value. It is also clear that the less reliable the proxy for the managers' effort is, the worse off the shareholders will be (i.e., the more distant their solution is from the first-best).

In conclusion, the perfect knowledge of the action would destroy the agency costs associated with the monitoring of the management.

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15. Optimality of golden parachutes

Source: Tirole, Ch. 7, Exercise 7.9, p. 326 (PDF 339) · Answer in book: yes, pp. 604–605

Exercise 7.9 (optimality of golden parachutes). Return to the manipulation model of Section 7.2.1, with the possibility of informed manipulation. Confirm the heuristic analysis of that section through a careful analysis, allowing for general contracts (the reward R_b^r or R_b^q is contingent on the revealed information and may *a priori* exceed $B/\Delta p$; a fixed payment can be made in both states and only under revealed poor prospects: L_l^r and $L_l^q \leqslant L$; allow $q_H R$ to be larger or smaller than L).

Context needed to solve it *From Tirole §3.2.1–3.2.2 (the fixed-investment model), pp. 115–117 (PDF 128–130):*

[…] an entrepreneur […] has a project. This project requires a fixed investment I. The entrepreneur initially has "assets" ("cash on hand" or "net worth") $A < I$. […] To implement the project the entrepreneur must borrow $I - A$ from lenders. […] If undertaken, the project either succeeds, that is, yields verifiable income $R > 0$, or fails, that is, yields no income. […] Behaving yields probability $p = p_H$ of success and no private benefit to the entrepreneur, and misbehaving results in probability $p = p_L < p_H$ of success and private benefit $B > 0$ (measured in units of account) to the entrepreneur. […] Let $\Delta p \equiv p_H - p_L$. […] Both the borrower and the potential lenders (or "investors") are risk neutral. […] the rate of return expected by investors (which is also the riskless rate, due to risk neutrality) is taken to be 0. The borrower is protected by limited liability, and so her income cannot take negative values. Lenders behave competitively in the sense that the loan, if any, makes zero profit. […]

[…] the borrower will therefore behave if the following "incentive compatibility constraint" is satisfied:

$$p_H R_b \geqslant p_L R_b + B \quad \text{or} \quad (\Delta p) R_b \geqslant B. \qquad (\text{IC}_b)$$

From Tirole §7.2.1.1 "Managerial Myopia: The Incentive for Posturing", pp. 301–303 (PDF 314–316):

Consider the fixed-investment model of Section 3.2, with the new ingredient that there is some learning at an intermediate stage about the entrepreneur's ability to run the project and the concomitant opportunity to replace her (or to liquidate the project) on the basis of this information (see Figure 7.7). The manager's type, which is a synonym for the probability of success, is either $r_\bullet$ or $q_\bullet$, where the dot subscript refers to the fact that the probability of success is not solely determined by the manager's ability and is a function of later effort (high or low).

Figure 7.7 (timeline, left to right): Funding stage. Entrepreneur invests I, of which $I-A$ is borrowed from investors. → Manipulation? → Probability of success ($r_\bullet$ or $q_\bullet$) learned publicly. → Liquidation/replacement (yields L), otherwise continue → Moral hazard (high or low effort). → Outcome (success: R, failure: 0).

No manipulation. Let us for the moment rule out any managerial manipulation of the intermediate information received by the investors. The latter learn at the intermediate stage that the probability of success in the case of continuation $\pi_\bullet = (\pi_H, \pi_L)$ (that is, contingent on effort: π_H in the case of good behavior, π_L in the case of misbehavior) is either high ($r_\bullet = (r_H, r_L)$) or low ($q_\bullet = (q_H, q_L)$), with $r_H > q_H$ and $r_L > q_L$. At the funding stage, no one knows which prevails, and the prior on the two possibilities is $(\alpha, 1-\alpha)$:

$$\pi_\bullet = \begin{cases} r_\bullet & \text{with probability } \alpha, \\ q_\bullet & \text{with probability } 1-\alpha. \end{cases}$$

[...] For simplicity, we will also assume that the manager's "type" ($r_\bullet$ or $q_\bullet$) is orthogonal to later moral hazard. So, if r_H and q_H denote the probabilities of success in the case of good behavior, and r_L and q_L those in the case of misbehavior, then

$$r_H - r_L = q_H - q_L = p_H - p_L = \Delta p,$$

letting $p_H \equiv \alpha r_H + (1-\alpha) q_H$ and $p_L \equiv \alpha r_L + (1-\alpha) q_L$ refer to the prior beliefs that the project will succeed under good and bad behavior, respectively. Thus, regardless of the manager's type, shirking reduces the probability of success by Δp.

Suppose that, when it accrues, the information about the manager's type is public and, for simplicity, verifiable; and that, contingent on the signal, the initial contract specifies whether management is allowed to continue or not. In the case of termination, the firm generates an expected profit L that can be shared between investors and incumbent management.

Example. Suppose that the manager's replacement is another similar manager of unknown ability in the job. Then, the "liquidation" value is

$$L = p_H \left(R - \frac{B}{\Delta p} \right).$$

For, the new management must be provided with a reward, $B/\Delta p$, in the case of success that induces the high effort. The pledgeable income is therefore equal to $p_H[R - B/\Delta p]$.

We make the following assumption:

$$q_H R > L. \qquad (7.9)$$

Inequality (7.9) says that, ceteris paribus, even a low-ability manager would prefer to keep her job, as this yields a higher NPV than termination. Put differently, it is *ex ante* efficient for the entrepreneur to retain her job. [...]

We will also assume that

$$I - A > \max \left\{ p_H \left(R - \frac{B}{\Delta p} \right), L \right\}, \qquad (7.10)$$

while

$$I - A < \alpha r_H \left(R - \frac{B}{\Delta p} \right) + (1-\alpha) L. \qquad (7.11)$$

The first inequality (7.10) states that guarantee of either tenure (the entrepreneur always keeps her job) or termination (the entrepreneur is always fired) does not generate enough pledgeable income to attract investors. Because the incentive problem is independent of the entrepreneur's type, the entrepreneur must be rewarded at least $B/\Delta p$ for

success in order to have an incentive to behave. And so the pledgeable income under guaranteed tenure is insufficient to cover the investors' initial outlay, $I - A$. By contrast, the second inequality, (7.11), which requires that

$$q_H\left(R - \frac{B}{\Delta p}\right) < L < r_H\left(R - \frac{B}{\Delta p}\right),$$

implies that there is enough pledgeable income to attract investors when there is termination in case of low ability, provided that the investors receive the return L in the case of termination.

Under a competitive capital market the entrepreneur's utility in case of funding is equal to the NPV:

$$U_b(z^r, z^q) = \alpha[z^r(r_H R) + (1 - z^r)L] + (1 - \alpha)[z^q(q_H R) + (1 - z^q)L] - I,$$

where z^r and z^q are the contracted-for probabilities of continuation of employment of a high- and low-ability entrepreneur, respectively. From (7.9), this utility is maximized by a guaranteed tenure:

$$z^r = z^q = 1.$$

Guaranteed tenure, however, does not attract funds (from (7.10)); and so some (contingent) termination must be conceded in order to satisfy the investors' breakeven constraint:

$$\alpha\left[z^r r_H\left(R - \frac{B}{\Delta p}\right) + (1 - z^r)L_l^r\right] + (1 - \alpha)\left[z^q q_H\left(R - \frac{B}{\Delta p}\right) + (1 - z^q)L_l^q\right] \geqslant I - A,$$

where L_l^r and L_l^q ($\leqslant L$) are the lenders' returns in the case of termination of a high- and low-ability manager, respectively. Clearly, setting

$$L_l^r = L_l^q = L$$

is optimal since this relaxes the investors' breakeven constraint without altering the NPV. Also, it is more efficient (in terms of maximizing both the NPV and the pledgeable income) to retain a high-ability manager:

$$z^r = 1.$$

Let $z^q = z^*$. From (7.10) and (7.11), the value $z^* \in (0, 1)$ is the smallest value that satisfies the investors' breakeven constraint:

$$\alpha r_H\left(R - \frac{B}{\Delta p}\right) + (1 - \alpha)\left[z^* q_H\left(R - \frac{B}{\Delta p}\right) + (1 - z^*)L\right] = I - A. \qquad (7.12)$$

Some termination in case of low ability is the concession made by the entrepreneur to attract investors. As is familiar, the entrepreneur sacrifices value (NPV) to boost pledgeable income.

Manipulation. Until now, we have assumed that the information received by the investors at the intermediate stage lies outside the entrepreneur's control. Let us now assume that the entrepreneur can, at a cost, alter this information.

More precisely, suppose that the entrepreneur can generate the high signal $r_\bullet$ by (secretly) manipulating the information. This manipulation comes at a cost: the probability of success falls (uniformly) by $\tau > 0$.

We distinguish two forms of manipulation:

Uninformed manipulation. The entrepreneur does not know her type when deciding whether to manipulate information (so, she learns her type at the same time as the investors in Figure 7.7).

Informed manipulation. The entrepreneur knows her type when choosing whether to manipulate the information (but learns it only after the funding stage, which therefore still occurs under symmetric information).

Suppose, in the first step, that the financing contract specifies, besides the probabilities of continuation z^r and z^q for signals $r_\bullet$ and $q_\bullet$, a reward $R_b \geqslant B/\Delta p$ in the case of continuation and success (and no payment to the entrepreneur otherwise). [...]

From Tirole §7.2.1.1, pp. 303–304 (PDF 316–317):

Under *informed manipulation*, the entrepreneur is tempted to manipulate the information only when she learns that she is inefficient. The new no-manipulation constraint is

$$z^r[(q_H - \tau)R_b] \leqslant z^q[q_H R_b]$$

or

$$\frac{z^r}{z^q} \leqslant \frac{1}{1 - \tau/q_H}. \qquad (7.14)$$

[...] The relevant no-manipulation constraint, (7.13) in the case of uninformed manipulation and (7.14) in the case of informed manipulation, will be labeled (NM).

Whether manipulation is uninformed or informed, the initial contract must in general lower z^r or increase z^q—in a nutshell, make continuation less signal dependent—or both. [...]

The threat of manipulation may prevent the firm from receiving funding in the first place. Start from the solution ($z^r = 1$, $z^q = z^*$) as defined by equation (7.12) when there is no scope for manipulation, and suppose that the ratio $1/z^*$ does not satisfy the relevant (NM) constraint. That is, [...]

$$\frac{1}{z^*} > \frac{1}{1 - \tau/q_H}$$

under informed manipulation.

If one keeps $z^r \equiv 1$, then z^q must be increased above z^* so as to satisfy the (NM) constraint. Increasing z^q above z^*, however, is not feasible since this reduces pledgeable income, which then becomes insufficient to cover the investors' initial outlay. Thus continuation cannot be guaranteed to a high-ability entrepreneur ($z^r < 1$).

This reduction in z^r reduces pledgeable income as $r_H[R - B/\Delta p] > L$. Hence, z^q must also be brought down below z^* in order to make up for the shortfall in pledgeable income.

As one could have expected, *the entrepreneur's ability to cook the books ex post may jeopardize funding ex ante.* And, even if funding is feasible, this ability reduces the NPV.

From Tirole §7.2.1.2 "Golden Parachutes", pp. 304–305 (PDF 317–318):

Top managers often receive very large compensation packages when their employment is terminated. These "golden parachutes" appear particularly "obscene" when termination is motivated by poor performance. Of course, many of these packages result from the board being in cahoots (or not wanting to enter any conflict) with top management. There is also some efficiency rationale for golden parachutes. Intuitively, the "softened landing" that they offer to managers makes them less prone to engage in various venal behaviors, such as earnings manipulations, in order to keep their job. In a nutshell, proponents of golden parachutes argue that they are the price to pay for incentive compatibility.

Are golden parachutes beneficial here? They are clearly costly as they reduce pledgeable income. However, a golden parachute helps relax the (NM) constraint. In a sense, they create more "balanced" incentives for the manager by increasing her payoff in the case of liquidation. Indeed, consider (for example) the case of informed manipulation and suppose that the entrepreneur receives some amount $T \geqslant 0$ when admitting that prospects are poor, i.e., when the signal is $q_\bullet$. The new (NM) constraint is

$$z^r[(q_H - \tau)R_b] \leqslant z^q[q_H R_b] + T, \qquad (7.15)$$

where $R_b = B/\Delta p$ in order to maximize the income that can be pledged to investors.

The key question is whether it is cheaper to prevent manipulation by making tenure relatively insensitive to new information or by granting a golden parachute ($T > 0$). To answer this question, let us write the NPV (which does not depend on T),

$$U_b(z^r, z^q, T) = \text{NPV} = \alpha[L + z^r(r_H R - L)] + (1-\alpha)[L + z^q(q_H R - L)] - I,$$

and the pledgeable income,

$$\mathcal{P}(z^r, z^q, T) = \alpha[L + z^r[r_H(R - R_b) - L]] + (1-\alpha)[L + z^q[q_H(R - R_b) - L] - T] = I - A.$$

Intuitively, there are two "currencies" available for paying the manager: continuation and golden parachute. A golden parachute is just a cash transfer while the continuation policy affects the NPV. One would therefore expect a golden parachute to be used exactly when the continuation policy is an inefficient policy, that is, when continuation under poor prospects reduces the NPV.

To demonstrate this "efficient currency result," suppose that $T > 0$. Looking at the pledgeable income, a unit increase in z^q (which increases the *ex ante* utility U_b because of the assumption that $q_H R > L$, but reduces pledgeable income), must be compensated by a decrease in the golden parachute equal in absolute value to

$$\left|\frac{dT}{dz^q}\right| = L - q_H(R - R_b).$$

From (7.15), this marginal change that keeps investor income constant relaxes the (NM) constraint:

$$dz^q(q_H R_b) + dT = dz^q[q_H R_b - L + q_H(R - R_b)] = (q_H R - L)\, dz^q > 0.$$

Thus the optimal golden parachute policy is to have none:

$$T^* = 0.$$

Not so when continuation under poor prospects reduces the NPV, and not only the investor income. Suppose now that

$$q_H R < L.$$

Then, from the previous reasoning, a golden parachute is a cheaper instrument than an insensitive tenure to keep the entrepreneur from manipulating accounts. When it is optimal to fire the manager in the case of poor prospects, she is paid a golden parachute:

$$z^{q*} = 0 \quad \text{and} \quad T^* = z^r(q_H - \tau)R_b > 0.$$

Exercise 7.9 asks the reader to check this heuristic reasoning more formally.

From Tirole §7.2.1.2, footnotes 51–53, p. 305 (PDF 318):

51. This compensation is slightly different from a golden parachute, since the latter would be received contingent on termination (in particular, when $z^r < 1$, the golden parachute would be received with positive probability even though the manager does not admit to poor prospects). The form of golden parachute considered here is more efficient because it is more effective at addressing the (NM) constraint. But it relies on our assumption that the state of nature is contractible. The analysis would not change much with the alternative formalization.

52. R_b is assumed to be the same in both states, but this involves no loss of generality.

53. So, in the previous notation, $L_l^r = L$ and $L_l^q = L - T/(1 - z^q)$.
